

Problem Set I

Political Science 43401

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1. If sets A and B are disjoint, what do I know about x if I know that $x \in B$? Give a concrete example.

- 3.1.1 For each of the following pairs of sets A and B , state which, if either, is a subset of the other:

- (a) A is the set of people living in Scotland; B is the set of people living in the UK.
- (b) A is the set of people living in the USA; B is the set of people living in California.
- (c) A is the set of natural numbers divisible by 6; B is the set of natural numbers divisible by 2.
- (d) A is the set of natural numbers divisible by 5; B is the set of natural numbers divisible by 7.

- 3.1.3 Let A, B, C be subsets of a set X . Which of the expressions $(A \cap B) \cup (A \cap B^c)$, $(A \cap B) \cup (A \cap B^c)^c$, $(A \cap B) \cup (A^c \cup C)^c$ can be simplified, and how?

- 3.4.2 Let g be the mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (y, -x),$$

and let the mapping h be as in Exercise 3.4.1. Find the image of (x, y) under each of the composite mappings $g \circ h$ and $h \circ g$.

- 31.1.1 As in the text, let $\mathbb{Q}$ be the set of all rational numbers. If $x \in \mathbb{Q}$, which of the following statements (P, Q, R, S) imply which of the others?

P: $x \geq 0$

Q: $\exists w \in \mathbb{Q}$ such that $x = w^2$

R: $nx > -1 \forall n \in \mathbb{N}$

S: $\exists n \in \mathbb{N}$ such that $nx > -1$.

R Bonus Using R, plot the following functions:

$$y(x) = \ln(x) + 4$$

$$y(x) = \frac{1}{2}e^{x^2/2}$$

$$z(x, y) = 3x^5y^5$$